leetcode.com /problems/merge-intervals/description/

Unknown Title

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56. Merge Intervals

Medium

Topics

Companies

Given an array of `intervals` where `intervals[i] = [start`$_i$`, end`$_i$`]`, merge all overlapping intervals, and return *an array of the non-overlapping intervals that cover all the intervals in the input*.

Example 1:

```
Input: intervals = [[1,3],[2,6],[8,10],[15,18]]
Output: [[1,6],[8,10],[15,18]]
Explanation: Since intervals [1,3] and [2,6] overlap, merge them into [1,6].
```

Example 2:

```
Input: intervals = [[1,4],[4,5]]
Output: [[1,5]]
Explanation: Intervals [1,4] and [4,5] are considered overlapping.
```

Example 3:

```
Input: intervals = [[4,7],[1,4]]
Output: [[1,7]]
```

Explanation: Intervals [1,4] and [4,7] are considered overlapping.

Constraints:

- $1 <= intervals.length <= 10^4$
- $intervals[i].length == 2$
- $0 <= start_i <= end_i <= 10^4$

Seen this question in a real interview before?

1/5

Yes

No

Accepted

3,932,468/7.6M

Acceptance Rate

51.5%

Topics

ArraySorting

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Discussion (286)

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💡 Discussion Rules

1. Please don't post **any solutions** in this discussion.

2. The problem discussion is for asking questions about the problem or for sharing tips - anything except for solutions.

3. If you'd like to share your solution for feedback and ideas, please head to the solutions tab and post it there.

leet_leak

Mar 31, 2023

The list is not sorted! Please give a clarification!

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1.1K

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Jatin Bedi

Jan 02, 2024

in given examples it is giving assumption that array is sorted.Please provide better test cases leetcode team

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Ayush kumar

Jun 17, 2024

This question broke all my confidence

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Aditya Kumar

Jun 02, 2024

imp test case:-
[[1,3],[2,6],[8,10],[15,18]]
[[1,4],[4,5]]

[[1,4],[0,0]]
[[1,3], [2,6] ,[8,10] ,[8,9], [9,11],[15,18], [2,4] ,[16,17]]

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Michael Camele

Oct 10, 2021

Title.

Am I missing something in the explination of overlap?

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Jonathan Janetzki

Jan 24, 2020

The problem description does not state that all interval starts are less or equal than their ends. However, this is true for all test cases.

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Lan W

Oct 22, 2018

This is a follow-up question in interviews. Does anyone have any idea?

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want to discuss about the FB follow up question, "How do you add intervals and merge them for a large stream of intervals? (Facebook Follow-up Question)" from here, https://leetcode.com/problems/merge-intervals/discuss/355318/Fully-Explained-and-Clean-Interval-Tree-for-Facebook-Follow-Up-No-Sorting

If this is an real world abstrcted problem, i'd prefer to use divide and conquer technique depends on the requirements. For example, partitioning input intervals based on interval start/end value, i.e we have 1000

buckets, inside each bucket we maintain a sorted intervals, each bucket can be in one machine or multiples buckets in one machine. then we use merge those bucket 2 by 2 until to the point each bucket contains maximum intervals. And make sure no intervals are across two buckets

For either endless incoming stream mentioned here or memory is too small to load all intervals, this approach should work.
this BST approach is impresive but not realistic for me to write in the interview https://leetcode.com/problems/merge-intervals/discuss/21451/Share-my-BST-interval-tree-solution-C%2B%2B-No-sorting!

Hope this can help, thanks

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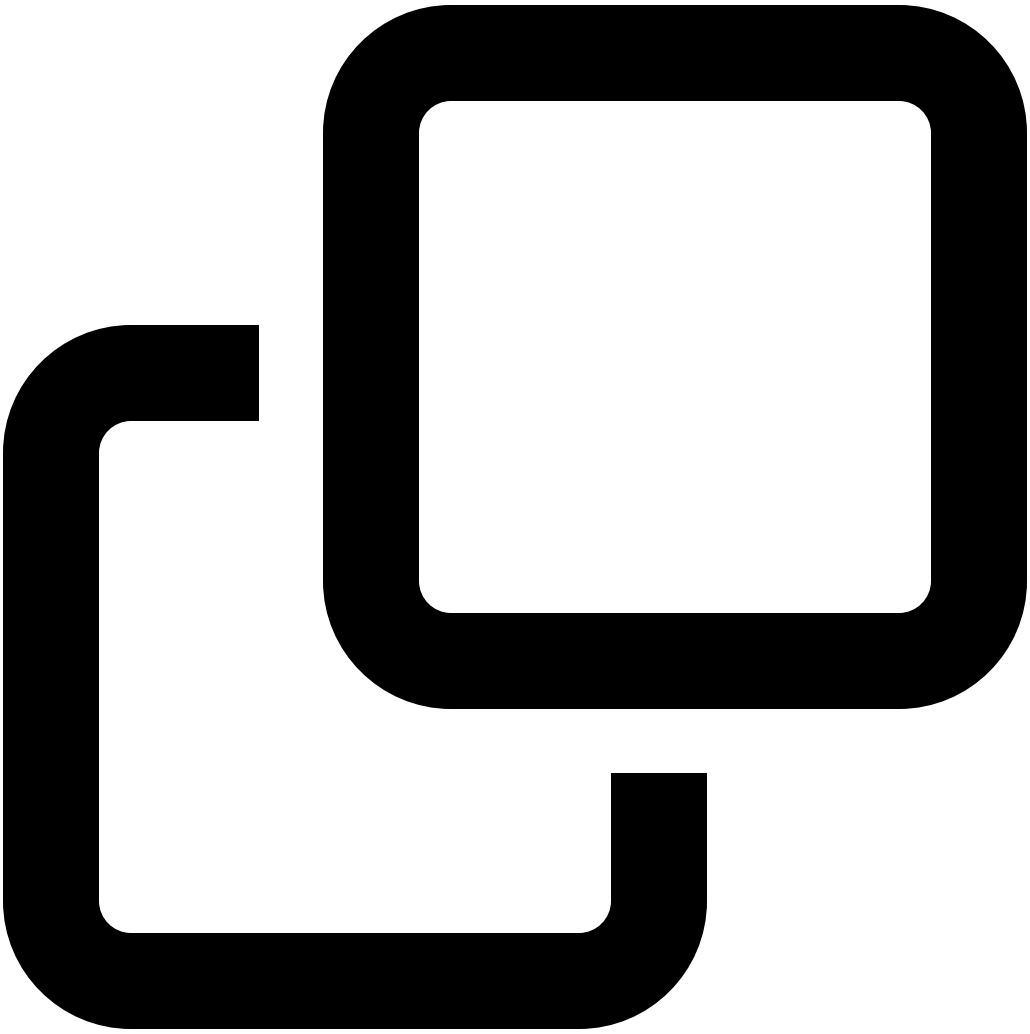
Yang

Oct 15, 2024

Test cases

```
[[1,3],[2,6],[8,10],[15,18]]
[[1,4],[4,5]]
[[1,1],[2,2],[0,0],[2,3],[1,3],[3,5],[2,3],[3,5]]
[[2,3],[2,2],[3,3],[1,3],[5,7],[2,2],[4,6]]
[[1,3],[2,6],[8,10],[8,9],[9,11],[15,18],[2,4],[16,17]]
[[1012,1136],[1137,1417],[1015,1020]]
[[1,3]]
[[1,4],[2,3]]
```



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How do they expect us to return an array in result when we wouldn't even know exactly how many intervals we are going to have??

```
public int[][] merge(int[][] intervals)
```

If you are expecting us to change List to array then why not just make the return List itself.

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Feedback

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